

Shaft design project report

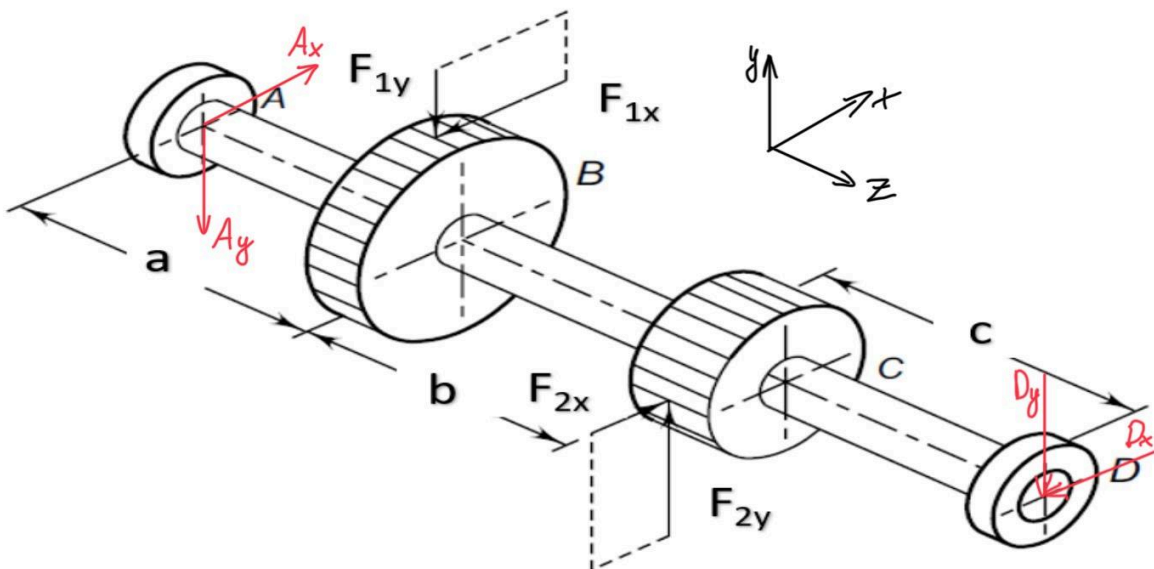
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Topic 14

1. Given parameters:

N [kW]	15
a [mm]	120
b [mm]	150
c [mm]	120
n [rpm]	800
m [mm]	8
Z_1	13
Z_2	40
β_1	0°
α (pressure angle)	20°

2. Scheme



3. Material for the shaft

In this project, I chose St4 Steel (approximately equivalent to C15), which is suitable for components with moderate mechanical requirements. It offers good machinability and weldability, making it ideal for parts that are not heavily stressed. St4 Steel is commonly used for smaller diameter shafts, axles, pins, and light-duty mechanical components.

4. Calculation of forces acting on the shaft

Twisting moment:

$$T = \frac{15 \times 10^3}{800 \times \frac{2\pi}{60}} = 180 Nm$$

Pitch diameters:

$$d_1 = 13 \times \frac{0.008}{\cos(0^\circ)} = 0.104 m$$

$$d_2 = 40 \times \frac{0.008}{\cos(0^\circ)} = 0.32 m$$

Circumferential Forces:

$$F_{t1} = \frac{2 \times 180}{0.104} = 3.461 kN$$

$$F_{t2} = \frac{2 \times 180}{0.32} = 1.125 kN$$

Radial Forces:

$$F_{r1} = 3.461 \times \frac{\tan(20^\circ)}{\cos(0^\circ)} = 1.26 kN$$

$$F_{r2} = 1.125 \times \frac{\tan(20^\circ)}{\cos(0^\circ)} = 0.41 kN$$

Axial Forces:

$$F_{o1} = 3.461 \times \tan(0^\circ) = 0 N$$

$$F_{o2} = 1.125 \times \tan(0^\circ) = 0 N$$

5. Calculation of Reaction Forces

$$y: \Sigma M = 0 \Rightarrow -a \times F_{1y} + (a + b)F_{2y} - (a + b + c)D_y = 0 \Rightarrow$$

$$D_y = \frac{0.12 \times 410 - 0.27 \times 3461}{-0.39} = 2.269 kN$$

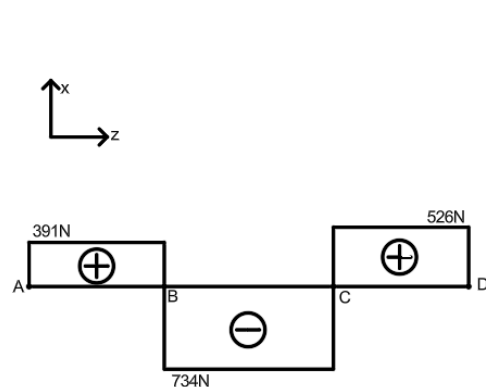
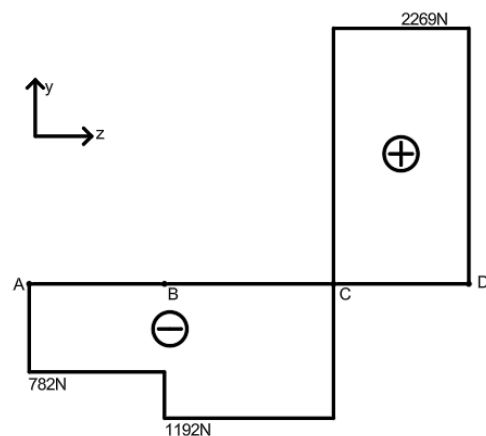
$$x: \Sigma M = 0 \Rightarrow -a \times F_{1x} + (a + b)F_{2x} - (a + b + c)D_x = 0 \Rightarrow$$

$$D_x = \frac{0.12 \times 1125 - 0.27 \times 1260}{-0.39} = 0.526 kN$$

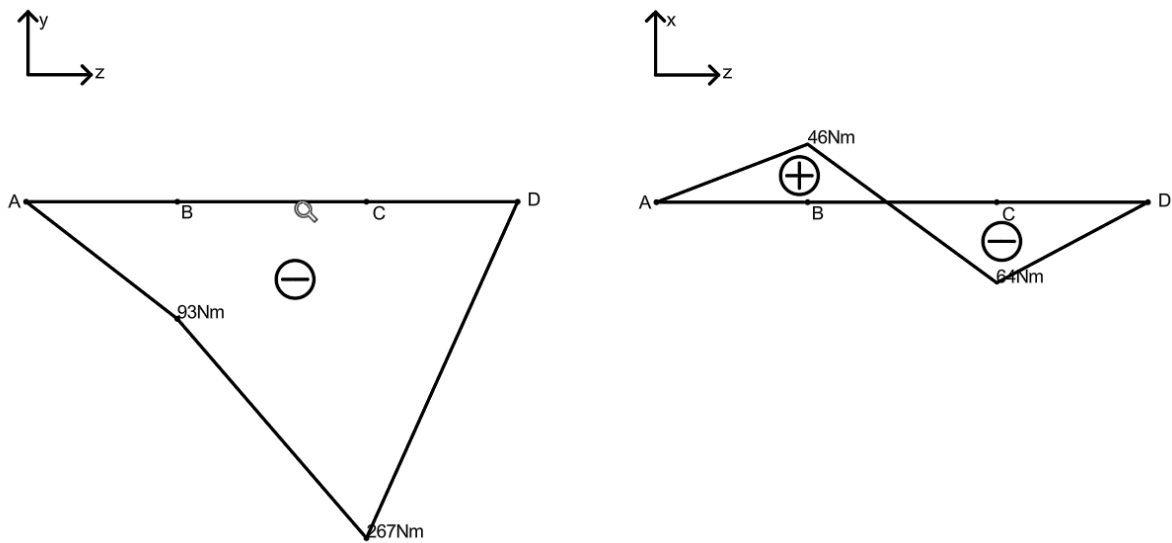
$$A_y = 0.782 kN, A_x = 0.391 kN$$

6. Shear Force and Moment diagrams

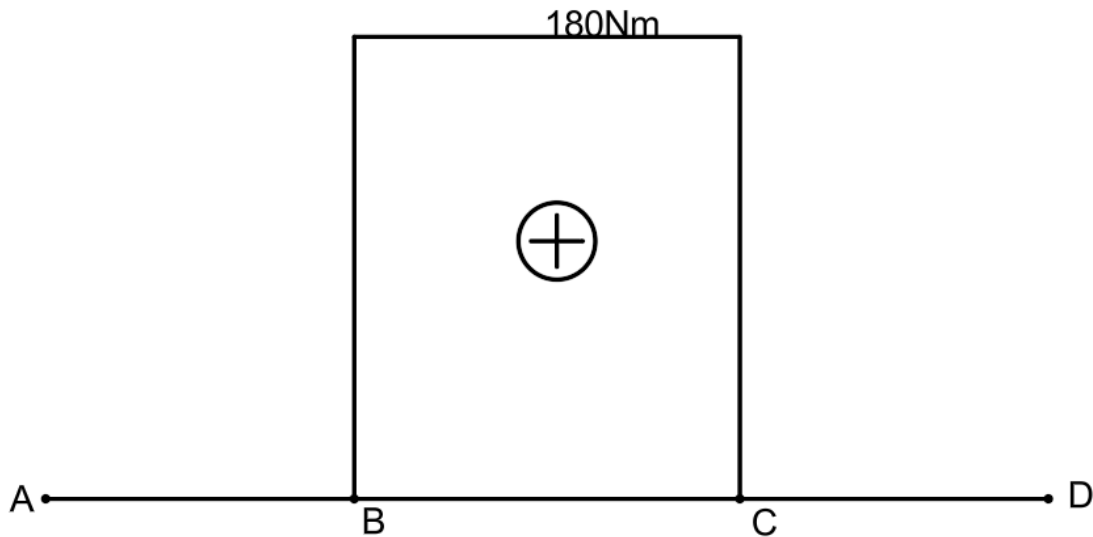
Shear Force diagrams:



Bending Moment diagrams:



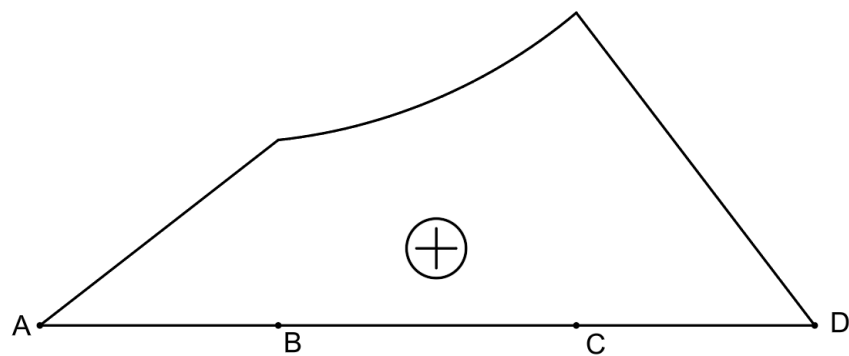
Twisting Moment diagram:



7. Equivalent moment calculations and diagram

$$\text{At point B: } M_{eq} = \sqrt{103.75^2 + \frac{3}{4} \times 180^2} = 187.25 \text{ Nm}$$

$$\text{At point C: } M_{eq} = \sqrt{274.56^2 + \frac{3}{4} \times 180^2} = 315.73 \text{ Nm}$$



8. Calculation of resultant moment at different points

For this point, we chose 6 different new points. The set location for those points will be measured as a distance from point A:

- Point E: 50 mm
- Point F: 100 mm
- Point G: 170 mm
- Point H: 220 mm
- Point I: 320 mm
- Point J: 370 mm

For each of those points, we will calculate resultant moments

For point E:

$$M_y = A_y \times 0.05 \text{ m} = 39.1 \text{ Nm}$$

$$M_x = A_x \times 0.05 \text{ m} = 19.55 \text{ Nm}$$

$$M_E = \sqrt{39.1^2 + 19.55^2} = 43.72 \text{ Nm}$$

$$M_{Eeq} = 43.72 \text{ Nm}$$

For point F:

$$M_y = A_y \times 0.1 \text{ m} = 78.2 \text{ Nm}$$

$$M_x = A_x \times 0.1 \text{ m} = 39.1 \text{ Nm}$$

$$M_F = \sqrt{39.1^2 + 78.2^2} = 87.43 \text{ Nm}$$

$$M_{Feq} = 87.43 \text{ Nm}$$

For point G:

$$M_y = A_y \times 0.17 + F_{1y} \times (0.17 - 0.12) = 153.44 \text{ Nm}$$

$$M_x = A_x \times 0.17 + F_{1x} \times 0.05 = 10.22 \text{ Nm}$$

$$M_G = \sqrt{153.44^2 + 10.22^2} = 153.78 \text{ Nm}$$

$$M_{Geq} = \sqrt{153.78^2 + \frac{3}{4} \times 180^2} = 218.97 \text{ Nm}$$

For point H:

$$M_y = A_y \times 0.22 + F_{1y} \times (0.22 - 0.12) = 213.04 \text{ Nm}$$

$$M_x = A_x \times 0.22 - F_{1x} \times 0.1 = -26.48 \text{ Nm}$$

$$M_H = \sqrt{213.04^2 + 26.48^2} = 214.68 \text{ Nm}$$

$$M_{Heq} = \sqrt{214.68^2 + \frac{3}{4} \times 180^2} = 265.31 \text{ Nm}$$

For point I:

$$M_y = A_y \times 0.32 + F_{1y} \times (0.32 - 0.12) - F_{2y} \times (0.32 - 0.27) = 159.19 \text{ Nm}$$

$$M_x = A_x \times 0.32 - F_{1x} \times 0.2 + F_{2y} \times 0.05 = -36.88 \text{ Nm}$$

$$M_I = \sqrt{159.19^2 + 36.88^2} = 163.41 Nm$$

$$M_{Ieq} = 163.41 Nm$$

For point J:

$$M_y = A_y \times 0.37 + F_{1y} \times (0.37 - 0.12) - F_{2y}(0.37 - 0.27) = 45.74 Nm$$

$$M_x = A_x \times 0.37 - F_{1x} \times 0.25 + F_{2y} \times 0.1 = -10.58 Nm$$

$$M_I = \sqrt{45.74^2 + 10.58^2} = 46.95 Nm$$

$$M_{Ieq} = 46.95 Nm$$

And now, for each point, we will calculate the allowable diameter using this formula:

$$d = \sqrt[3]{\frac{32}{\pi \times \sigma_{allow.}} \times M_{eq}}, \text{ where } \sigma_{allow.} = 150 \text{ MPa}$$

$$E: d = \sqrt[3]{\frac{32}{\pi \times \sigma_{allow.}} \times M_{Eq}} = 0.014 m = 14 \text{ mm}$$

$$F: d = \sqrt[3]{\frac{32}{\pi \times \sigma_{allow.}} \times M_{Feq}} = 0.018 m = 18 \text{ mm}$$

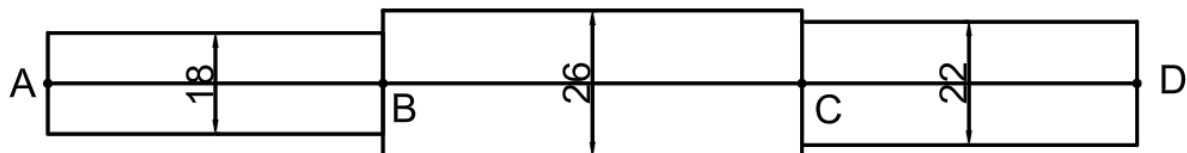
$$G: d = \sqrt[3]{\frac{32}{\pi \times \sigma_{allow.}} \times M_{Geq}} = 0.024 m = 24 \text{ mm}$$

$$H: d = \sqrt[3]{\frac{32}{\pi \times \sigma_{allow.}} \times M_{Heq}} = 0.026 m = 26 \text{ mm}$$

$$I: d = \sqrt[3]{\frac{32}{\pi \times \sigma_{allow.}} \times M_{Ieq}} = 0.022 m = 22 \text{ mm}$$

$$J: d = \sqrt[3]{\frac{32}{\pi \times \sigma_{allow.}} \times M_{Jeq}} = 0.015 m = 15 \text{ mm}$$

After all calculations, we can draw a silhouette of our shaft:



9. Designing step shaft

In this point we chose the roller bearings, keys, splines and gears:

For keys:

Re = 280 MPa - yield strength

Rm = 400 MPa

$$b = \frac{d}{4} \quad b = \frac{2}{3b} = \frac{d}{6} \quad l = 1.5 * d$$

$$H: b = \frac{26}{4} = 6.5 mm \quad h = \frac{2}{3} * 6.5 = 4.33 mm \quad l = 1.5 * 26 = 39 mm$$

$$\tau = 2 * \frac{180}{0.0065 * 0.039 * 0.026} = 54.62 \text{ MPa}$$

$$\delta = \frac{4 * 180}{0.00433 * 0.039 * 0.026} = 163.99 \text{ MPa}$$

$$K_r = \frac{280 \text{ MPa}}{2} = 140 \text{ MPa}$$

$$Zrc = 0.7 * 140 = 98$$

In the end i chose keys again basing on DIN 6885 standard - A 7x7x40
Steel C45

Splines:

$$D\Delta r = \frac{D + d}{2} = \frac{38 + 26}{2} = 32 \text{ mm}$$

We choose 10 - tooth spline

$$p = \frac{11250}{10 * 2.8 * 39.1} = 10.3 \text{ MPa}$$

10 - tooth involute spline (conservative, common for 26 - 30mm shafts)

Bearing SKF 6003 (18mm bore)

Bore: 18mm

Outer d: 35mm

Width: 10mm

Dynamic Load Rating C: 4.55kN

Static Load Rating C: 1.95kN

Max speed (greased): 19 000 rpm

Load 1.26kN < 4.55 kN

Pin:

$$\text{Force on pin: } F = \frac{2T}{\pi d^2} = 13846 \text{ N}$$

$$\tau = \frac{2F}{\pi d^2} \leq \tau_{allow}$$

$$\frac{2 * 13846}{\pi d^2} \leq 90 \Rightarrow d^2 \geq 97.88$$

$$d = 9.89 \text{ mm}$$

Choice: 10 mm steel pin DIN EN ISO 8734, h6fit

Pitch diameters: $m = 5\text{mm}$, $z = 20\text{mm}$, $d = 100\text{mm}$

$$F_t = \frac{2T}{d} = \frac{2 * 180}{0.1} = 3600\text{N} = 3.6\text{kN}$$

Check Bending Strength of Gear:

$$F_t \leq 100 * 50 * 5 * 0.35 = 8.75\text{kN}$$

According to the book, we've chosen gear:

Type: Spur Gear

Module: 5

Wraith: 50mm

Pressure angle: 20degrees

Number of teeth: 20

10. Checking stiffness of the shaft

each part of the shaft gives us a different moment of inertia:

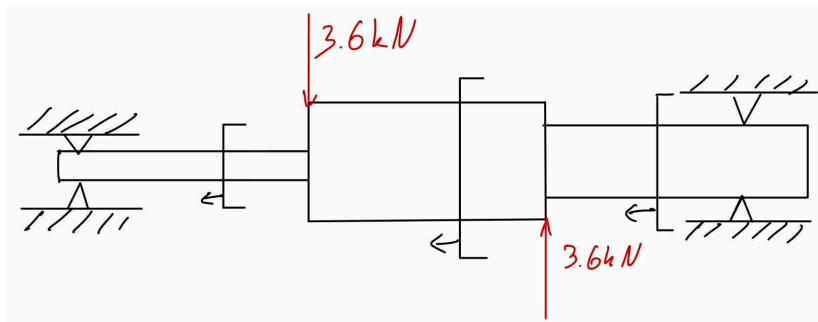
$$I_{AB} = \frac{\pi * d^4}{64} = 5.15 * 10^{-9}\text{m}^4$$

$$I_{BC} = \frac{\pi * d^4}{64} = 2.24 * 10^{-8}\text{m}^4$$

$$I_{CD} = \frac{\pi * d^4}{64} = 1.15 * 10^{-8}\text{m}^4$$

$$E = 2.1 * 10^{11}\text{Pa} = 200\text{GPa}$$

11. Calculation of deflections



For M_1 ($0 < x < 120$):

$$M_y = 0$$

For M_1 ($120 < x < 270$):

$$M_y(x) = 0x|_{AB} + 3600(x - 0.12)|_{BC}$$

$$-EI_y W''(x) = 3600x - 432$$

$$-EI_y W'(x) = 1800x^2 - 432x + C_1$$

$$-EI_y W(x) = 600x^3 - 216x^2 + C_1x + C_2$$

$$-EI_y = 4480$$

$$W(x) = -0.13392x^3 + 0.04821x^2 + \frac{1}{4480}C_1x + \frac{1}{4480}C_2$$

$$W(0) \Rightarrow 0 = C_2$$

$$0 = -0.13392(270)^3 + 0.04821(270)^2 + \frac{1}{4480}C_1 \times 270 + 0$$

$$C_1 = 4679806$$

$$W(120) = 10\mu m$$

For M_3 ($270 < x < 390$):

$$M_y(x) = 0x|_{AB} + 3600(x - 0.12)|_{BC} - 3600(x - 0.27)|_{CD}$$

$$W''(x) = \frac{-M_y(x)}{EI_y}$$

$$M_y(x) = 0 + 3600x - 432 - 3600x + 972$$

$$M_y(x) = 0 + 540$$

$$-EI_y W''(x) = 540$$

$$-EI_y W'(x) = 540x + C_1$$

$$-EI_y W(x) = 270x^2 + C_1x + C_2$$

The boundary condition to calculate the deflection of a beam at two points.

$$-EI_y = 2300$$

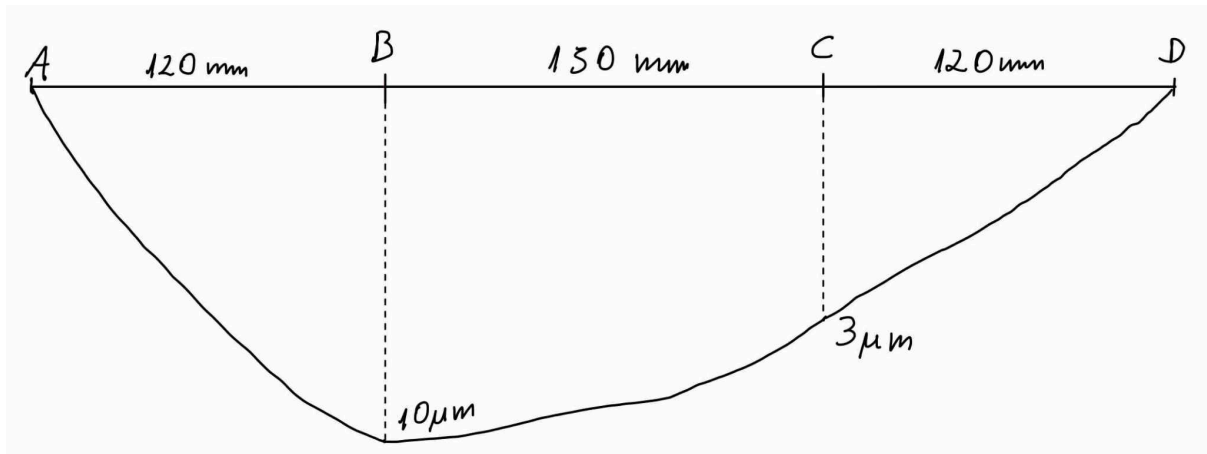
$$W(x) = 0.1173913x^2 + \frac{1}{2300}C_1x + \frac{1}{2300}C_2$$

$$W(0) \Rightarrow C_2 = 0$$

$$W(390) \Rightarrow C_1 = -105298$$

$$0 = 0.11739(390)^2 + \frac{390}{2300}C_1 + 0$$

$$W(270) = 0.11739(270)^2 - \frac{105298}{2300} \times 270 = 3\mu m$$



12. Calculation of critical speed

$$\omega_c = \frac{60}{2\pi} \sqrt{\frac{g \sum_{i=1}^n \omega_i \delta_i}{\sum_{i=1}^n \omega_i \delta_i^2}}, \text{ where } g = 9.81 \text{ m/s}^2, \text{ and } \omega_1 = \omega_2 = 180 \text{ N}$$

$$\text{Numerator: } g \sum_{i=1}^n \omega_i \delta_i = 229.878$$

$$\text{Denominator: } \sum_{i=1}^n \omega_i \delta_i^2 = 1.962$$

$$\text{Ratio under square root: } \frac{229.878}{1.962} = 117.17$$

$$\text{Final answer: } \omega_c = 10,366 \text{ rpm}$$

12. Calculations for fatigue strength

For these calculations, we will use the following parameters:

Z_{go}	150 MPa
Z_{so}	160 MPa
M_{g1}	103.75 MPa

M_{g2}	274.56 MPa
T	180 MPa
β_p	1.3
ε	0.62
α_k	2.6
η_k	0.80
D	100 mm
R_e	280 MPa

Now we can proceed with the calculations for the first point:

$$\beta_k = 1 + \eta_k(\alpha_k - 1) = 1 + 0.8 \times 1.6 = 1 + 1.28 = 2.28$$

$$\beta = \beta_p + \beta_k - 1 = 2.58$$

$$\sigma_{n1} = \frac{32 \times M_{g1}}{\pi \times D^3} = 1.06 MPa$$

$$\delta_{\sigma 1} = \frac{\frac{Z_{go}}{\beta} \times \varepsilon}{\beta \times \sigma_{n1}} = 34.01 MPa$$

$$\tau_a = \frac{8 \times T}{\pi \times D^3} = 0.458 MPa$$

$$\delta_{\tau 1} = \frac{Z_{so}}{\tau_a \times \left(\frac{\beta}{\varepsilon} + \frac{Z_{so}}{R_e} \right)} = 73.81 MPa$$

$$\delta = \frac{\delta_{\sigma 1} \times \delta_{\tau 1}}{\sqrt{\delta_{\sigma 1}^2 + \delta_{\tau 1}^2}} = 30.89 MPa$$

And for point 2:

$$\beta_k = 1 + \eta_k(\alpha_k - 1) = 1 + 0.8 \times 1.6 = 1 + 1.28 = 2.28$$

$$\beta = \beta_p + \beta_k - 1 = 2.58$$

$$\sigma_{n2} = \frac{32 \times M_{g2}}{\pi \times D^3} = 2.8 MPa$$

$$\delta_{\sigma 2} = \frac{\frac{Z_{go}}{\beta} \times \varepsilon}{\beta \times \sigma_{n2}} = 12.87 MPa$$

$$\tau_a = \frac{8 \times T}{\pi \times D^3} = 0.458 MPa$$

$$\delta_{\tau 2} = \frac{Z_{so}}{\tau_a \times \left(\frac{\beta}{\varepsilon} + \frac{Z_{so}}{R_e} \right)} = 73.81 MPa$$

$$\delta = \frac{\delta_{\sigma 2} \times \delta_{\tau 2}}{\sqrt{\delta_{\sigma 2}^2 + \delta_{\tau 2}^2}} = 12.68 MPa$$

Below are the assembly and manufacturing drawings of the shaft:

